Preparing Dicke States on Willow

The mathematical choices behind our solution

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Our function `create_dicke_state` returns an exact circuit of CZ and `PhasedXZ` gates on calibrated qubits of Google's Willow chip. The submission is `submission.ipynb`, the challenge notebook with `main.py` embedded verbatim. This note states the mathematical fact behind each design choice, with a short justification for each. Section 12 covers the bonus track.

We assume linear algebra (vectors, 2×2 matrices), first-year probability (expected value, variance, the normal distribution) and one-variable calculus. Every quantum notion is defined where it first appears. A gentler four-idea overview is in `docs/writeup/main.pdf`.

Every number here is *simulated* with calibrated gate errors, energy decay ($T_1 = 68\,\mu$s [3]), dephasing ($T_\varphi = 100\,\mu$s, assumed) and asymmetric readout errors. Nothing here has run on hardware. File paths refer to the repository.

§1 Background and notation

§1.1 Qubits and the Dicke state

A **qubit** is a unit vector $a|0\rangle + b|1\rangle$ in $\mathbb{C}^2$, where $|0\rangle$ and $|1\rangle$ are the two standard basis vectors. Measuring it gives 0 with probability $|a|^2$ and 1 with probability $|b|^2$. A state of N qubits is a unit vector with one **amplitude** ψ_b for each of the 2^N bit strings b:

$$|\psi\rangle = \sum_{b\in\{0,1\}^N} \psi_b |b\rangle, \qquad \sum_b |\psi_b|^2 = 1.$$

Measuring all qubits gives the string b with probability $|\psi_b|^2$. We write $|b|$ for the **weight** of b, the number of ones in it. Every state in this note has real amplitudes.

Definition 1 (Dicke state)**.** The **Dicke state** $|D_n^N\rangle$ is the equal superposition of all N-bit strings of weight n:

$$|D_n^N\rangle = \frac{1}{\sqrt{K}} \sum_{|b|=n} |b\rangle, \qquad K = \binom{N}{n}.$$

The case $n = 1$ is the **W state** $|W_N\rangle$. We also write $k = \min(n, N-n)$.

Example 1 (D_2^4)
$|D_2^4\rangle = \frac{1}{\sqrt{6}}\big(|0011\rangle + |0101\rangle + |0110\rangle + |1001\rangle + |1010\rangle + |1100\rangle\big)$, with $K = \binom{4}{2} = 6$. Measuring gives each of the six strings with probability 1/6.

The circuit must produce this vector *exactly*, signs included, when there is no noise. A measurement only reveals the probabilities $|\psi_b|^2$, not the signs. That is why the judges also measure in a second basis (Section 2.2).

§1.2 Gates

A **gate** is a unitary matrix applied to one or two qubits. The ones we need are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$R_y(\theta) = e^{-i\theta Y/2} = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \qquad \mathrm{CZ} = \mathrm{diag}(1, 1, 1, -1).$$

These are the **Pauli matrices** X, Y, Z and the rotation generated by Y. X flips a bit, Z flips the sign of $|1\rangle$, and $R_y(\theta)$ rotates the real plane spanned by $|0\rangle, |1\rangle$ by the angle $\theta/2$. The **controlled-Z** gate CZ acts on two qubits and flips the sign of $|11\rangle$ only.

The rules allow any single-qubit gate (Cirq's `PhasedXZ` gate covers every 2×2 unitary), but the only two-qubit gate is CZ, and only between qubits that are physically connected on the chip. A CZ fails roughly ten times as often as a single-qubit gate, so a circuit's main cost is its **CZ count**. The second cost is its **depth**, the number of time steps (**moments**), because waiting qubits decay as well.

§1.3 The noise we simulate

Our simulator (`experiments/`) uses four kinds of error:

- **Gate errors.** After a gate with calibrated error rate e, with probability e a random Pauli error (X, Y or Z on the qubits involved) hits the state. This model is called **depolarizing noise**.
- **Energy decay.** An idle qubit in $|1\rangle$ relaxes to $|0\rangle$ at rate $1/T_1$.
- **Dephasing.** An idle qubit in a superposition picks up a random sign between $|0\rangle$ and $|1\rangle$ at rate $1/T_\varphi$.
- **Readout errors.** Each qubit reads 1 as 0 with probability $e_q^{1\to0}$ and 0 as 1 with probability $e_q^{0\to1}$. The first rate is typically about twice the second.

§2 The judges' score

Each test case (N, n) is judged from 10^4 shots in the Z basis (measure every qubit directly) and 10^4 in the X basis (explained below). The score of the case is

$$S = \frac{1}{z + x},$$

and the total is the sum of S over all test cases. The pairs (N, n) are not disclosed.

§2.1 The Z term: is every valid string equally likely?

Throw away the Z-basis shots whose weight is not n. Let f be the fraction that remain, let $F = 10^4 f$ be their number, and let c_b count the shots that gave string b. If the state

were perfect, each of the K valid strings would be expected F/K times. The judges compare the counts with this even split using Pearson's **chi-squared statistic**:

$$\chi^2 = \sum_{|b|=n} \frac{(c_b - F/K)^2}{F/K}, \qquad z = \frac{\chi^2}{Kf}.$$

So z is small when the counts look evenly split and when few shots were thrown away.

§2.2 The X term: are the signs right?

Measuring in the X **basis** means applying a Hadamard gate to every qubit before reading it out. A bit b_i read this way is the outcome of the observable X_i, whose value is $(-1)^{b_i}$. For a set A of qubits, the product $X_A = \prod_{i\in A} X_i$ has value $(-1)^{|b\cap A|}$, where $|b \cap A|$ counts the ones of b inside A. Averaging over shots estimates the **correlator** $\langle X_A\rangle = \langle\psi|X_A|\psi\rangle$.

Let $\bar{C}_j$ be the average of $\langle X_A\rangle$ over all $\binom{N}{j}$ subsets A of size j, and $C_j^\star$ its ideal value. Then

$$x = \sum_{j=1}^{M} \varepsilon_j^2, \qquad \varepsilon_j = \begin{cases} (\bar{C}_j - C_j^\star)/C_j^\star, & C_j^\star \neq 0, \\ \bar{C}_j, & C_j^\star = 0. \end{cases}$$

M is undisclosed. We report $M = N$, the strictest choice.

Proposition 2 (Ideal correlators)

For $|D_n^N\rangle$, $C_j^\star = \binom{j}{j/2}\binom{N-j}{n-j/2} \Big/ \binom{N}{n}$ when j is even, and $C_j^\star = 0$ when j is odd.

Proof. X_A flips the bits in A, so $X_A|s\rangle = |s \oplus A\rangle$. Then

$$\langle D_n^N|X_A|D_n^N\rangle = \frac{1}{K}\sum_{|s|=n}\sum_{|t|=n}\langle t|s\oplus A\rangle,$$

and the inner sum is 1 exactly when $s \oplus A$ again has weight n. If s has i ones inside A, flipping A changes the weight by $j - 2i$, so the weight is kept exactly when $i = j/2$. That is impossible for odd j. For even j, the number of such s is $\binom{j}{j/2}$ (choose which half of A is ones) times $\binom{N-j}{n-j/2}$ (place the other ones outside A). □

Example 2

For $|D_2^4\rangle$: $C_2^\star = \binom{2}{1}\binom{2}{1}/6 = 2/3$ and $C_4^\star = \binom{4}{2}/6 = 1$. The second value says that flipping all four bits maps $|D_2^4\rangle$ to itself, which is the complement symmetry of Section 4.

§2.3 Fast scoring

Evaluating x directly means averaging over all $\binom{N}{j}$ subsets for every shot. A counting argument removes that cost.

Claim — For one X-basis shot b of weight w, the average of $(-1)^{|b\cap A|}$ over all

j-subsets A is $\mathcal{K}_j(w)/\binom{N}{j}$, where $\mathcal{K}_j(w) = \sum_i (-1)^i \binom{w}{i}\binom{N-w}{j-i}$ is a **Krawtchouk polynomial**.

Proof. A subset A with i of its j elements among the w ones of b contributes $(-1)^i$. There are $\binom{w}{i}\binom{N-w}{j-i}$ such subsets. □

So x depends only on the histogram of shot weights, not on the strings themselves. That is what makes 20,000 simulated judged runs per case cheap (`experiments/scoring.py`).

§3 The score is heavy-tailed: small K and uniformity come first

The score is the *reciprocal* of a random quantity. If z is usually about 1 but occasionally lands very close to 0, then S is usually about 1 but occasionally enormous. Those rare large values dominate the average. This section makes that precise.

§3.1 The chi-squared distribution

Definition 3. If $G_1, \dots, G_\nu$ are independent standard normal variables, the distribution of $G_1^2 + \cdots + G_\nu^2$ is the **chi-squared distribution with ν degrees of freedom**, χ^2_ν. Its density on $t > 0$ is

$$p_\nu(t) = \frac{t^{\nu/2-1} e^{-t/2}}{2^{\nu/2}\,\Gamma(\nu/2)},$$

where Γ is the gamma function, $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\,dt$, which satisfies $\Gamma(s+1) = s\,\Gamma(s)$.

Fact 4. For a perfect state and many shots, Pearson's statistic χ^2 follows χ^2_{K-1}.

Each term $(c_b - F/K)/\sqrt{F/K}$ is approximately normal, by the central limit theorem. The K counts must add up to F, which removes one independent direction, so $K-1$ degrees of freedom remain.

Lemma 5
$\mathbb{E}[1/\chi^2_\nu] = 1/(\nu-2)$ for $\nu > 2$, and $\mathbb{E}[1/\chi^2_\nu] = \infty$ for $\nu \le 2$.

Proof. Near $t = 0$ the integrand $p_\nu(t)/t$ behaves like $t^{\nu/2-2}$, which is integrable exactly when $\nu/2 - 2 > -1$, that is, $\nu > 2$. In that case, the integral $\int_0^\infty t^{\nu/2-2} e^{-t/2}\,dt$ is the same as in the normalization of $p_{\nu-2}$, so

$$\mathbb{E}[1/\chi^2_\nu] = \frac{2^{(\nu-2)/2}\,\Gamma(\nu/2-1)}{2^{\nu/2}\,\Gamma(\nu/2)} = \frac{1}{2\,(\nu/2-1)} = \frac{1}{\nu-2}.$$ □

§3.2 Expected score of a perfect state

With f and x held fixed, $S = 1/\big(\chi^2/(Kf) + x\big) = Kf/(\chi^2 + a)$ with $a = Kfx$. So

$$\mathbb{E}[S] = Kf\,\mathbb{E}\left[\frac{1}{\chi^2_{K-1} + a}\right] \approx \begin{cases} \dfrac{Kf}{K-3}, & K > 3, \\ \frac{3}{2} f\, e^{a/2} E_1(a/2) \;\approx\; \frac{3}{2} f\big(\ln\frac{2}{a} - \gamma\big), & K = 3, \\ 2f\sqrt{\pi/(2a)}\ \operatorname{erfcx}\sqrt{a/2} \;\approx\; \sqrt{\pi f/x}, & K = 2. \end{cases}$$

Here $E_1(u) = \int_u^\infty e^{-t}/t\,dt$ is the **exponential integral**, $\gamma \approx 0.577$ is Euler's constant, and $\operatorname{erfcx}(u) = e^{u^2}\operatorname{erfc}(u)$ is the scaled complementary error function. Both of the last two are close to their small-a forms because a is tiny.

- $K > 3$. This follows from Lemma 5 with $\nu = K - 1 > 2$, ignoring the small a. The mean is finite even at $x = 0$.
- $K = 3$. Here $p_2(t) = \frac{1}{2}e^{-t/2}$, and substituting $u = (t + a)/2$ gives $\int_0^\infty \frac{e^{-t/2}}{2(t+a)}\,dt = \frac{1}{2}e^{a/2}E_1(a/2)$ exactly. For small u, $E_1(u) \approx -\ln u - \gamma$.
- $K = 2$. Here $p_1(t) = e^{-t/2}/\sqrt{2\pi t}$. Substituting $t = u^2$ turns the integral into the standard form $\int_0^\infty e^{-u^2/2}/(u^2 + a)\,du$, which gives the erfcx expression exactly. Since $\mathrm{erfcx}(0) = 1$, the score grows like $x^{-1/2}$ as $x \to 0$.

For $K \leq 3$, Lemma 5 says the mean would be infinite at $x = 0$. Only the X term keeps it finite.

K	2	3	4	6	10	20
$\mathbb{E}[S]$, perfect state, $f = 0.9$, $x = 10^{-3}$	51.4	8.15	3.34	1.79	1.28	1.06

At $K = 2$: $x = 10^{-4}$ gives 166 and $x = 10^{-2}$ gives 15.2 (`scoring.expected_score`).

A single $K = 2$ case, such as W_2, is worth about fifty large cases.

§3.3 Imbalance removes the tail

Suppose valid string b appears with probability $(1 + \delta_b)/K$ inside the subspace instead of $1/K$. Then each term of χ^2 is a squared normal with a nonzero mean. Such a sum follows the **noncentral chi-squared distribution**, whose **noncentrality** λ is the sum of the squared means:

$$\lambda = \sum_b \frac{\big(F(1 + \delta_b)/K - F/K\big)^2}{F/K} = F\,\overline{\delta^2} = 10^4 f\,\overline{\delta^2},$$

where $\overline{\delta^2}$ is the average of δ_b^2 over the K strings. The mean becomes $\mathbb{E}[\chi^2] = K - 1 + \lambda$, so z grows by about $10^4\,\overline{\delta^2}/K$. More importantly, χ^2 can no longer land near 0, so the lucky draws disappear.

Example 3

A 1% rms imbalance ($\delta_{\mathrm{rms}} = 0.01$, so $\lambda \approx 0.9$) lowers $\mathbb{E}[S]$ at $K = 2$ from 51.4 to 33.3, and a 3% imbalance ($\lambda \approx 8$) lowers it to 1.3. At $K = 4$ the same steps give 3.34, 2.53 and 0.53.

Remark 6 (Consequences)**.** What this means for the design:

- The judges sum single draws, so our objective is $\sum \mathbb{E}[S]$. We compute it by integrating $1/(z + x)$ against the noncentral χ^2 density numerically, with x held at its mean. Averaging resampled runs does not work at $K \leq 3$, where a few lucky draws dominate the average.
- Priority order: a valid circuit on every case, since any valid circuit scores above 0; then uniformity ($\delta_{\mathrm{rms}} \lesssim 1\%$) and a low x floor at small K; then the fraction f of valid shots, which costs only $1/f$; then coherence at large N.
- We may not pre-compensate known biases by tilting the amplitudes, because the noiseless state must be exact. Uniformity can only come from the construction and from the choice of qubits.

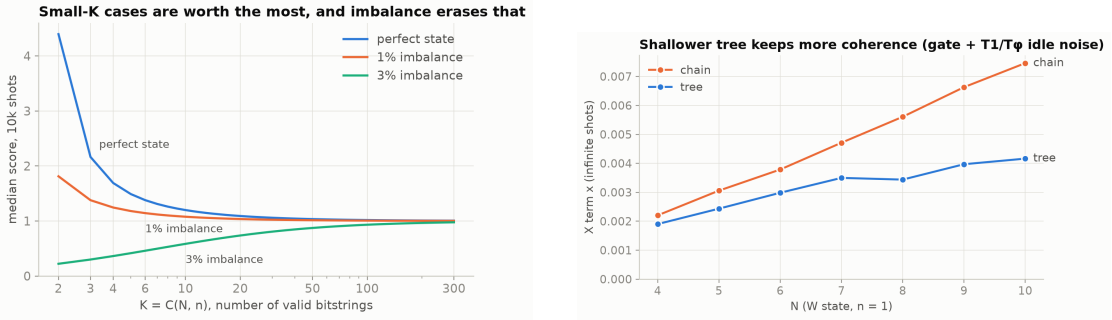


Figure 1: Left: median single-draw score versus K for a perfect state and for 1% and 3% rms imbalance. Right: X term of W_N under gate and idle noise, tree versus chain (`w_tree.csv`, Section 5).

§3.4 The organizers' revised preliminary ranking

On October 25 the organizers changed the preliminary leaderboard. It now averages $1/S$ over trials and reports $1/\overline{1/S} = 1/\mathbb{E}[z+x]$, because the plain mean diverges as the noise vanishes (Lemma 5). With $\mathbb{E}[\chi^2] = K - 1 + \lambda$,

$$\frac{1}{\mathbb{E}[z+x]} = \frac{1}{\dfrac{K-1+\lambda}{Kf} + x} \approx \frac{1}{\dfrac{1}{f} + \dfrac{\lambda}{Kf} + x} \qquad (K \gg 1).$$

This removes the heavy tail. The organizers tested D_3^{10}, D_5^{10} and D_4^{15}, with $K = 120$, 252 and 1365. For those the score is set by the total error budget through f and by coherence through x; imbalance enters only as λ/K. On that ranking our first submission led on D_3^{10} and trailed the teammates' preliminary combined submission on D_5^{10} and D_4^{15}. The error budgets traced the gap to dense single-qubit pulses (Section 9). The notebook still describes final judging as one draw per case on hardware, so $\sum \mathbb{E}[S]$ stays our objective and we report $1/\mathbb{E}[z+x]$ next to it.

§4 Two free reductions

Complement. $X^{\otimes N}|D_n^N\rangle = |D_{N-n}^N\rangle$, because flipping every bit maps weight-n strings one-to-one onto weight-$(N-n)$ strings. So we only build $n = k \le N/2$, and the final layer of X gates merges into the last single-qubit gates at no cost.

Permutation symmetry. Swapping two qubits maps $|D_n^N\rangle$ to itself, and the returned qubits are a set, not an ordered list. So any placement of a construction's shape on the chip is a valid answer, and qubit order never has to be restored after routing.

§5 W states ($k = 1$): a broadcast tree with $2N - 3$ CZ

A W state has a single excitation (one "1") shared evenly among N qubits. Think of it as probability 1 that must be split into N equal shares of $1/N$.

§5.1 Construction

Algorithm — Choose a spanning tree of the chosen qubits with root r, and let $s(v)$ be the number of qubits in the subtree under v (including v). Start with r in $|1\rangle$ and the rest in $|0\rangle$. In each round, every qubit v that currently holds probability

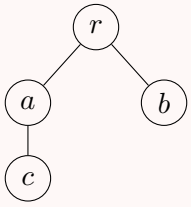
m_v/N hands one child c the fraction $p = s(c)/m_v$ of it, using the two-qubit **transfer**

$$|1\rangle_v|0\rangle_c \mapsto \sqrt{1-p}\,|10\rangle + \sqrt{p}\,|01\rangle, \qquad |00\rangle \mapsto |00\rangle.$$

Afterwards c holds $s(c)/N$, exactly what its subtree needs, and v keeps $(m_v - s(c))/N$. By induction every qubit ends with exactly $1/N$. Every branch of the superposition carries one excitation, and all amplitudes stay real and positive, so the output is exactly $|W_N\rangle$.

Example 4 (W_4 in two rounds)

Take the four-qubit tree below, rooted at r.



Round 1: r holds everything and gives a the fraction $p = s(a)/4 = 1/2$. Round 2, both at once: r gives b half of its share, and a gives c half of its share. Every qubit now holds $1/4$. A line $r - a - c - b$ would need three rounds.

§5.2 Cost

The very first transfer acts on a known product input $|1\rangle_v|0\rangle_c$. An R_y rotation on c gives $|1\rangle(\sqrt{1-p}|0\rangle + \sqrt{p}|1\rangle)$, and a CNOT from c to v (one CZ between Hadamards) turns it into $\sqrt{1-p}|10\rangle + \sqrt{p}|01\rangle$. That costs 1 CZ.

Every later transfer must act correctly on both $|00\rangle$ and $|10\rangle$, because v holds the excitation only in some branches. A gate that rotates within the span of $|01\rangle, |10\rangle$ and leaves $|00\rangle, |11\rangle$ alone is called a **Givens rotation**. It costs 2 CZ, and one is not enough:

Claim 7 — No circuit with a single CZ performs the transfer with $0 < p < 1$.

Proof. First, a CZ applied to a product state $(\alpha|0\rangle + \beta|1\rangle)|\phi\rangle$ gives $\alpha|0\rangle|\phi\rangle + \beta|1\rangle Z|\phi\rangle$. That is again a product state exactly when $\alpha\beta = 0$ or $Z|\phi\rangle \propto |\phi\rangle$. In words: exactly when one of the two inputs is $|0\rangle$ or $|1\rangle$ up to a phase.

A one-CZ circuit has the form $(A_1 \otimes B_1)\,\mathrm{CZ}\,(A_0 \otimes B_0)$, and the outer single-qubit gates $A_1 \otimes B_1$ cannot create or remove entanglement. The transfer must keep $|00\rangle$ a product state but entangle $|10\rangle$. Before the CZ, these inputs have become $A_0|0\rangle \otimes B_0|0\rangle$ and $A_0|1\rangle \otimes B_0|0\rangle$. If $B_0|0\rangle$ is a basis state, both stay products. Otherwise $A_0|0\rangle$ must be a basis state, which forces the orthogonal vector $A_0|1\rangle$ to be one too, and again both stay products. Either way $|10\rangle$ is not entangled, a contradiction. (Bryan's argument, `docs/bryan_constructions.md`.) □

The total is $1 + 2(N-2) = 2N - 3$ CZ. An exhaustive search shows this is optimal for W_3 (3 CZ) and W_4 (5 CZ).

§5.3 Why a tree

A chain (line) has the same CZ count but $N - 1$ sequential transfers. The tree runs the transfers of a round in parallel, so the number of rounds grows roughly like $\log_2 N$. W_{10}

takes 15 moments instead of 37, and under idle noise x drops from 0.0075 to 0.0042. The circuit also preserves the number of excitations, so every branch spends the same time in $|1\rangle$. Energy decay then lowers every valid string equally and creates no imbalance.

§5.4 The smallest cases

For W_2 ($K = 2$) the score is set by x and λ, not by CZ count (Section 3). We simulate every candidate qubit pair under the calibrated noise and keep the one with the highest predicted $\mathbb{E}[S]$. That gives +9% expected score and −5% median score, a deliberate trade toward the objective.

§6 Whole-state synthesis for $N \leq 8$

The general construction (Section 7) uses gates between qubits that are not neighbours on the chip, and fixing that adds many CZs. For small N we let a computer search for an exact circuit instead.

§6.1 The template

The circuit alternates $L + 1$ layers of R_y rotations, one angle per qubit per layer, with L layers of CZ gates. The CZs come from a fixed shape on the chip: a path, or a $2 \times m$ patch of the grid (a ladder). The edges of the shape are split into **matchings**, sets of edges that share no qubit, so the CZs of one layer can run at the same time. This is the **brickwork** pattern.

Why only R_y. Dicke amplitudes are real, and R_y and CZ are real matrices, so the search never leaves real vectors. The restriction costs no generality. Möttönen et al. [4] show that any real state can be prepared from R_y gates and CNOTs, and their CNOTs serve only to turn a rotation around via $XR_y(\theta)X = R_y(-\theta)$. The identity $ZR_y(\theta)Z = R_y(-\theta)$ does the same job, so each CNOT can be replaced by a CZ. Given enough connectivity, R_y and CZ therefore reach every real state.

§6.2 The search

Write $|\psi(\theta)\rangle$ for the output, which depends on the list of angles θ. The circuit is exact when the **infidelity**

$$\mathcal{L}(\theta) = 1 - \langle D_n^N | \psi(\theta) \rangle^2$$

is 0, since two unit vectors have overlap ± 1 only when they are equal up to sign. We minimize $\mathcal{L}$ with L-BFGS, a gradient method that builds an estimate of the curvature from recent gradients, until $\mathcal{L} < 10^{-13}$ (exact to machine precision). The gradient is computed exactly by backpropagation through the circuit, using

$$\frac{d}{d\theta} R_y(\theta) = \tfrac{1}{2} R_y(\theta + \pi),$$

which follows from $\cos(\frac{\theta}{2} + \frac{\pi}{2}) = -\sin\frac{\theta}{2}$ and $\sin(\frac{\theta}{2} + \frac{\pi}{2}) = \cos\frac{\theta}{2}$.

We start at the smallest brickwork depth that becomes exact. Then we delete single CZs greedily, re-solve from the previous angles, and keep a deletion only if the circuit becomes exact again (`experiments/synth.py`).

case	D_2^4	D_2^5	D_3^6	D_3^7	D_3^8	D_4^8
CZ, synthesized	5	10	15	18	30	26
CZ, routed B-E baseline	21	35	80	98	123	162

The D_3^7, D_3^8 and D_4^8 circuits come from our teammates' longer runs of the same template on Bridges-2 (more seeds, and a 2×4 patch for D_3^7 and D_3^8). They replaced our 25, 31 and 27 CZ and raise $1/\mathbb{E}[z + x]$ by 0.6–3.7% on three calibrations (`psc_hw*.csv`). Their 9-CZ D_2^5 and 14-CZ D_3^6 were not adopted: D_2^5 gained only 0.8% and would lose its robust angle pick (below), and D_3^6 measured 3% worse under idle noise.

No approximate shortcuts. Deleting any single CZ from a deployed pattern and refitting leaves an infidelity of 3×10^{-3} or worse for D_2^4, D_2^5, D_2^6 and D_3^6 (`vqe_gap.csv`). Approximate circuits are not allowed anyway, so fewer CZs can only come from a different pattern, which takes a new random start, not a deletion. Synthesis takes minutes to hours, so the angles are cached in `main.py`. Placement on the chip (8 grid symmetries $\times$ translations) happens at build time.

§6.3 Choosing among exact angle sets

One CZ pattern usually admits many exact angle sets. They behave the same under our noise model but differ under **coherent errors**, small systematic mistakes such as a rotation that is always 1% too large. Our noise model does not include these, so we analyse them separately.

> **Claim 8 —** In a real circuit, a small R_y over-rotation by ε moves the measured probabilities by $O(\varepsilon)$, but a stray Z phase or a CZ phase error moves them only by $O(\varepsilon^2)$.

Proof. Insert $e^{-i\varepsilon G} \approx I - i\varepsilon G$ somewhere in the circuit. The state changes from the real vector ψ to $\psi - i\varepsilon G\psi$ to first order.

- Over-rotation: $G \propto Y$, and $-iY$ is a real matrix. The change is real, so it adds directly to the real amplitudes and the probabilities ψ_b^2 move at first order.
- Phase errors: G is real and diagonal, so $-i\varepsilon G\psi$ is i times a real vector. Then $|\psi_b - i\varepsilon(G\psi)_b|^2 = \psi_b^2 + \varepsilon^2(G\psi)_b^2$, and the first-order term vanishes. □

On the deployed D_2^4, shrinking ε from 10^{-2} to 10^{-3} cuts the population error $10\times$ for over-rotation and $100\times$ for stray Z, as the claim predicts. So the choice only models over-rotation.

For a candidate angle set c, let λ_c be the imbalance it produces under a 1% over-rotation. Since $\delta \propto \sigma$ at first order, the imbalance scales like σ^2, so at over-rotation σ the expected loss is

$$L(c, \sigma) = \mathbb{E}[\text{loss}]_c + \lambda_c \, (\sigma/1\%)^2/(Kf),$$

by the $\lambda/(Kf)$ term of Section 3. We do not know σ, so we compare candidates by their **regret**, how much worse each is than the best candidate for that σ, and deploy the **minimax-regret** choice, whose worst-case regret over $\sigma \in \{0, 0.1, 0.2, 0.3, 0.5, 1\}\%$ is smallest:

$$c^\star = \arg\min_c \max_\sigma \Big[L(c, \sigma) - \min_d L(d, \sigma) \Big].$$

It is chosen on calibration seed 0 and checked on seeds 1–2. Sensitivity to 1% over-rotation drops 1.1–5.5$\times$ (D_2^5: λ from 50 to 9). Under our standard noise model, which has no over-rotation, the change in ceiling share (expected score as a percentage of a perfect state's) is between -0.9 and $+2.6$ points.

On the hardware path these picks are deployed for D_2^4, D_2^5 and D_3^6. (D_2^N with $N \geq 6$ goes to Bryan's circuits, Section 10.) The three adopted circuits D_3^7, D_3^8, D_4^8 run without one: each is a single exact angle set, and we have no alternatives on their structures to choose from.

§7 Bärtschi–Eidenbenz with synthesized SCS blocks

§7.1 The recursion

Sort the strings of $|D_r^m\rangle$ by their last bit. A fraction r/m of them end in 1, and the rest of such a string is a weight-$(r-1)$ string on $m-1$ bits. The others end in 0. So

$$|D_r^m\rangle = \sqrt{r/m}\,|D_{r-1}^{m-1}\rangle|1\rangle + \sqrt{(m-r)/m}\,|D_r^{m-1}\rangle|0\rangle.$$

Bärtschi and Eidenbenz [1] turn this identity into a circuit. Start from the sorted string $|0^{N-k}1^k\rangle$. The block SCS_m ("split and cyclic shift") decides the bit at position m with the right probability and leaves a sorted string on the first $m-1$ positions. Applying $\mathrm{SCS}_N, \mathrm{SCS}_{N-1}, \ldots, \mathrm{SCS}_2$ gives $|D_k^N\rangle$. Block SCS_m touches only the window of positions $m-k, \ldots, m$.

§7.2 Each block only needs to be right on a few inputs

Before step m, the first m positions hold a sorted string, so the window ($w = k+1$ qubits) only ever sees the inputs $|0^{w-l}1^l\rangle$ with $l = 0, \ldots, k$, plus $|1^w\rangle$ when the window is the whole register. On these inputs the block must act as

$$|0^{w-l}1^l\rangle \mapsto \sqrt{l/m}\,|0^{w-l}1^l\rangle + \sqrt{(m-l)/m}\,|0^{w-l-1}1^l0\rangle \quad (1 \le l \le k,\ l < m),$$

and fix $l = 0$ and $l = m$. Either the last bit stays 1 (probability l/m), or the block of ones shifts left and a 0 lands at position m. So each block is only an **isometry**: a length-preserving map specified on a $(k+1)$- or $(k+2)$-dimensional subspace, not a full unitary on all 2^{k+1} inputs. The rest of the matrix is free, which leaves room for cheaper circuits.

§7.3 Synthesis of the blocks

We fit each block on $k+1$ consecutive path qubits with the template and pruning of Section 6. With B input–output pairs $(\mathrm{in}_b, \mathrm{out}_b)$, the loss is

$$1 - \left(\tfrac{1}{B}\textstyle\sum_b \langle \mathrm{out}_b|U|\mathrm{in}_b\rangle\right)^2.$$

Each overlap is at most 1 in absolute value, so the loss vanishes only when every input is mapped correctly *with a common sign*. A separate sign per input would not be good enough: a superposition of inputs, which is what the block actually receives, would then come out with wrong relative signs.

For $k \le 4$ the CZ pattern is pruned once per k and the angles are refit for every m; at $k = 5$ some m got their own pattern. The whole B-E circuit is then nearest-neighbour on a path and needs no routing. Inner blocks cost 2, 4, 12 and 17 CZ for $k = 1, \ldots, 4$ and 22–29 for $k = 5$, so D_2^N costs about $4N$ CZ (`experiments/scs_synth.py`). Blocks exist for $m \le 20$. On the hardware track these circuits are only a fallback, for $9 \le N \le 20$ when Bryan's builder (Section 10) fails; beyond $N = 20$ the last resort is the routed B-E circuit.

§8 Placement: the cost is $\log \mathbb{E}[z]$

Willow's qubits and couplers differ a lot in quality, and by permutation symmetry (Section 4) a circuit can go anywhere its shape fits. So we score every placement and keep the best one.

§8.1 The cost of a placement

From Section 3, $\mathbb{E}[z] \approx (K-1+\lambda)/(Kf)$, so

$$\log \mathbb{E}[z] \approx \text{const} - \log f + \log\bigl(1+\lambda/(K-1)\bigr).$$

Approximate f by the probability that no error happens anywhere, $f \approx \prod(1-e)$ over all error locations. Since $\log(1-e) \approx -e$ for small e, $-\log f \approx \sum e$: the product becomes a sum, which is easy to optimize over. A placement's cost is therefore

$$C = \sum_{(q,q')} u_{qq'}\, e^{\mathrm{CZ}}_{qq'} + \sum_q \Bigl[g_q\, e^{1q}_q + \bigl(1-\tfrac{n}{N}\bigr)e^{0\to1}_q + \tfrac{n}{N}\, e^{1\to0}_q\Bigr] + \tau\Bigl(\tfrac{1}{T_1}+\tfrac{1}{T_\varphi}\Bigr) + \log\Bigl(1+\frac{\lambda_{\mathrm{ro}}}{K-1}\Bigr).$$

- $u_{qq'}$ is the number of CZs on coupler (q,q'), and $e^{\mathrm{CZ}}_{qq'}$ its error rate.
- g_q is the number of single-qubit gates on qubit q, and e^{1q}_q their error rate.
- Each qubit of $|D^N_n\rangle$ reads 1 with probability n/N, so its readout error is weighted accordingly. This matters because $1 \to 0$ errors are about twice as frequent as $0 \to 1$.
- τ is the total idle time. Decay over time τ has probability $1 - e^{-\tau/T} \approx \tau/T$.
- λ_{ro} is the imbalance caused by readout, derived next.

§8.2 Readout imbalance

A valid string b is read correctly with probability

$$\prod_{i\in b}(1-e^{1\to0}_i)\prod_{i\notin b}(1-e^{0\to1}_i) \approx c\,\exp\Bigl(-\sum_{i\in b} d_i\Bigr), \qquad d_i = e^{1\to0}_i - e^{0\to1}_i,$$

where $c = \prod_i(1-e^{0\to1}_i)$ does not depend on b. Strings whose ones sit on "leaky" qubits (large d_i) are undercounted. To first order, $\delta_b \approx -\sum_{i\in b}(d_i - \bar{d})$.

For a random choice of the n positions of the ones, this is a sum of n values drawn without replacement from the N values $d_i - \bar{d}$. Its variance is $\frac{n(N-n)}{N-1}\mathrm{Var}(d)$, the standard formula for sampling without replacement, so

$$\lambda_{\mathrm{ro}} = 10^4 f\,\frac{n(N-n)}{N-1}\,\mathrm{Var}(d)$$

(`main._readout_lambda`). A random layout gives $\delta_{\mathrm{rms}} \approx 0.6\%$. Since amplitudes cannot be pre-compensated, qubit choice is the only remedy.

§8.3 Winner's curse

The calibrated rates are noisy estimates. If we pick the placement with the lowest *estimated* cost among many, we tend to pick one whose errors happen to be underestimated, so the minimum is biased low. To counter this, every calibrated rate enters as value $+1\sigma$, its estimate plus one standard uncertainty. This changed none of 75 placements on the notebook's calibrations, where every qubit has the same uncertainty. It matters only when uncertainties differ between qubits.

§9 Compilation

Every moment of our circuits is either all-CZ or all-single-qubit.

§9.1 Single-qubit merge pass

Every circuit we return, Bryan's included (Section 10), goes through one final exact pass. On each qubit, a run of single-qubit gates between two of its two-qubit gates is replaced by the product of the run as one `PhasedXZ` gate, or dropped if that product is the identity up to a phase. The merged gate sits in the moment of the run's last gate, and empty moments are then removed.

The pass is exact because a run of gates and its matrix product are the same operation. No gate moves earlier, so no qubit leaves $|0\rangle$ sooner, and the moment count can only fall.

Why it matters. Each gate adds its own error. Bryan's constructions place a π pulse in every idle single-qubit slot (a dense echo sequence known as CPMG). At $N \geq 9$ these made up 60% of their single-qubit gates. Under depolarizing noise each pulse only adds its error rate, and on D_4^{15} the single-qubit error budget was as large as the CZ budget ($\sum e = 0.33$ against 0.27). The pass cuts D_4^{15} from 777 single-qubit gates to 276, D_5^{10} from 371 to 230 and D_4^9 from 286 to 170. Measured by $1/\mathbb{E}[z+x]$ on three calibrations, it gains 1–11% at $N \geq 9$ under gate noise, 0–21% under idle noise, and at most 1% at $N \leq 8$, where our own compiler already merged runs.

§9.2 Dynamical decoupling

On real hardware an idle qubit drifts: its relative phase between $|0\rangle$ and $|1\rangle$ rotates slowly by a random amount. Flipping the qubit with an X pulse makes the drift run backwards, so a well-timed sequence of pulses cancels it. This is **dynamical decoupling** (DD). We insert the sequence $XYXY$. Since $XY = iZ$, the sequence multiplies to $(iZ)^2 = -I$, which leaves the noiseless state unchanged up to a global phase.

We apply it only after a qubit's first gate, since $|0\rangle$ has no phase to lose, and only in idle windows with at least 4 free single-qubit slots. Short idles have nothing to refocus, so there only the pulse error would show. Dropping DD altogether scores another 1–5% in simulation. But our simulator's dephasing is memoryless, so it cannot show the slow drift that DD removes on hardware. The sparse echoes stay.

§10 Portfolio with Bryan's solution

Bryan's independent solution is vendored in `main.py`, and its derivations are in `docs/bryan_constructions.md`. Three ingredients:

- **A 1-CZ controlled rotation onto a $|0\rangle$ target.** Since $R_y(\frac{\theta}{2})ZR_y(-\frac{\theta}{2}) = R_y(\theta)Z$ and $Z|0\rangle = |0\rangle$, putting a CZ between $R_y(-\frac{\theta}{2})$ and $R_y(\frac{\theta}{2})$ on the target does nothing if the control is 0 and applies $R_y(\theta)$ if the control is 1. A general controlled rotation needs 2 CZ.
- **Nearest-neighbour SCS steps.** A 5-CZ "moving-hole" gate and a 4-CZ merged top block make each SCS step of Section 7 act on neighbours only.
- **A two-arm split.** Split the qubits into arms of sizes a and b with $a+b=N$. Then

$$|D_k^N\rangle = \sum_j \sqrt{P(j)}\,|D_{k-j}^a\rangle|D_j^b\rangle, \qquad P(j) = \binom{b}{j}\binom{a}{k-j}\Big/\binom{N}{k},$$

  because $P(j)$ is the fraction of weight-k strings with exactly j ones in the second arm (a **hypergeometric distribution**). Compare the divide-and-conquer circuits of [2].

The split value j is written in unary (as 1^j0^{k-j}, a "thermometer") by a chain of $k-1$ CZs and copied across a ladder with k CZs in one layer. Then both arms run B-E in parallel.

Under our harness it wins for $k=2$ at $N \geq 6$ (by 1–7%) and for $k \geq 3$ at $N \geq 9$ (21–39%; D_5^{10} needs 85 CZ against our 185), and `main.py` dispatches by that measured rule. Since October 25 his circuits also pass through the merge pass and our sparse DD (Section 9) instead of keeping his dense echoes.

§11 Results

case	K	construction	CZ	f	$\chi^2/(K-1)$	x	$\mathbb{E}[S]$	% of ceiling	$1/\mathbb{E}[z+x]$
W_2	2	tree, best pair	$2 \to 1$	0.990	1.041	0.0005	$54.26 \to 66.51$	$96 \to 96$	1.90
W_3	3	tree	$4 \to 3$	0.977	1.026	0.0013	$7.45 \to 8.03$	$92 \to 97$	1.43
D_2^4	6	synthesized	$21 \to 5$	0.968	1.019	0.0058	$1.55 \to 1.84$	$84 \to 95$	1.13
D_3^6	20	synthesized	$80 \to 15$	0.923	1.014	0.0347	$0.54 \to 1.03$	$48 \to 88$	0.93
D_3^8	56	synthesized (teammates')	$123 \to 30$	0.868	1.016	0.0958	$0.39 \to 0.83$	$38 \to 79$	0.80
D_4^8	70	synthesized (teammates')	$162 \to 26$	0.880	1.019	0.1113	$0.24 \to 0.82$	$24 \to 79$	0.80
D_2^{10}	45	Bryan's, merged	$97 \to 31$	0.850	1.016	0.0438	$0.38 \to 0.86$	$36 \to 81$	0.82
D_5^{10}	252	Bryan's, merged	$324 \to 85$	0.692	1.046	0.8695	$0.12 \to 0.42$	$12 \to 42$	0.42

How to read the table. Values are means over calibration seeds 0–2, with $M=N$, gate and idle noise, and 20,000 simulated runs per case. Arrows read "first-day baseline $\to$ deployed"; the baseline is routed B-E, with a chain for W states. The column $\chi^2/(K-1)$ is 1 for a perfectly balanced state, since $\mathbb{E}[\chi^2_{K-1}] = K-1$, so values above 1 measure imbalance. "% of ceiling" is $\mathbb{E}[S]$ as a percentage of the expected score of a perfect state.

Deployed values are a re-run of the final `main.py` (`wf_*.csv`), and the full 25-case table is in `ARCHITECTURE.md`. Over all 25 cases with $N \leq 10$ the re-run gives $\sum \mathbb{E}[S] = 102.0$ and $\sum$ median $= 28.2$, against 218.7 and 33.1 for a perfect state. The notebook benchmark of the final submission (`experiments/bench_notebook.py`, the notebook's own scorer) gives $\sum$ median $= 27.9$. W_2 and W_3 alone hold 74.5 of our expected total and 185.5 of the perfect-state total, which is why Section 3 drives the design.

The organizers' pairs under $1/\mathbb{E}[z+x]$. Before and after the merge pass, against the teammates' preliminary combined submission (gate noise / idle noise, three calibrations; D_4^{15} from 300 quantum trajectories on calibration seed 0):

case	before merge pass	deployed	preliminary combined
D_3^{10}	0.765 / 0.639	**0.776 / 0.639**	0.722 / 0.582
D_5^{10}	0.569 / 0.421	**0.603 / 0.421**	0.582 / 0.413
D_4^{15}	0.491 / 0.318	**0.546 / 0.351**	0.447 / 0.310

Our harness does not reproduce the size of the organizers' gaps. After the pass the deployed circuits lead under both noise presets on all ten cases from D_3^7 to D_4^{15} that we compared (`ARCHITECTURE.md`, Section 4).

> **Remark 9** (Limits)**.** Not modeled: coherent errors (studied separately in Section 6), ZZ crosstalk, leakage, per-qubit T_1 spread and drift. T_φ and layer durations are assumed. Real hardware will score lower.

§12 Bonus track: depth only

`create_dicke_state_adaptive` is scored without noise by

$$\sum_{2\le N\le 15,\ 1\le n\le N/2} \frac{Nn}{\text{moments}}$$

over 56 cases, so only depth matters. The organizers' first-round feedback put our entry at 37.15 against a leader at 41.10. All 56 cases were exact, so the gap was depth. Three facts fixed the design.

(i) The entry is unitary. The bonus track allows mid-circuit measurement and gates conditioned on the results (**feedforward**). But `Willow105.validate_circuit` rejects classically controlled operations; plain measurement and reset pass. A feedforward circuit would fail the validator the judges are told to use, and for $N \le 15$ every measurement-based scheme we costed was deeper anyway.

(ii) Where the depth is. Bryan's two-arm split has $2 \cdot (\text{CZ depth}) + 1$ moments (alternating CZ and single-qubit moments), and the CZ depth is set by the B-E sweep of the longer arm. Each arm position costs about 18 moments at $k \ge 5$ (10 at $k = 3$, 5 at $k = 2$). The sweeps of an arm's last few positions run on the same qubits with nothing in parallel: for D_7^{15}, 147 of 160 moments were the 8-qubit arm alone.

(iii) That tail is a small isometry. The last m_0 positions of a sweep implement $U_{m_0}\colon |0^{m_0-r}1^r\rangle \mapsto |D_r^{m_0}\rangle$ for $r \le \min(k, m_0)$. As for the SCS blocks (Section 7), only these inputs occur, so we synthesize U_{m_0} exactly with the same common-sign loss, fewest layers first (`experiments/tail_synth.py`). When m_0 equals the arm length, the whole arm is one block: U_7 with $r \le 2$ takes 27 moments against a 37-moment sweep, and U_6 with $r \le 4$ takes 43 against 97. The layer count found varies by 2–6 between random starts, so every block got 8–13 seeds.

The entry builds every exact candidate and returns the one with the fewest moments after packing single-qubit gates into CZ moments. The candidates are:

- our hardware constructions without DD, and Bryan's bonus circuits;
- two-arm splits with the thermometer on either arm and the last $m_0 \in \{3, \dots, 7\}$ positions (or the whole arm) replaced by U_{m_0};
- a teammate's staggered W tree, one to two moments shallower at $N \ge 5$;
- whole-state synthesized circuits ranked by moments rather than CZ, including the teammates' Bridges-2 circuits (D_2^7 in 17 moments, D_4^9 in 41).

Against the first-round entry, D_4^{12} drops from 88 to 51 moments, D_3^9 from 54 to 31, D_2^{15} from 46 to 35, D_7^{15} from 160 to 144 and W_9 from 14 to 12. `score_solution` gives **44.47** (`notebook_bonus.csv`); the teammates' own entry measures 41.47 with the same scorer. What remains is $k \ge 5$ (12 cases, 8.0 of the 44.47 points): those arms carry every weight up to the arm length, and no exact block beyond U_4 with $r \le 4$ was found for them.

§13 What did not work

- **Whole-state synthesis at $N = 9$–10.** An overnight run found exact circuits of 33, 35, 98 and 136 CZ for D_2^9, D_2^{10}, D_4^9 and D_3^9, against 27, 31, 63 and 50 deployed. Random-start brickwork stops scaling at $N \approx 9$.
- **Compressing Bryan's circuits.** Re-parametrizing his D_3^9 and deleting single CZs recovered exactness 0 of 8 times (best residual 1.6×10^{-4}).

- **ALAP scheduling** (starting every gate as late as possible) to keep qubits in $|0\rangle$ longer changed $\mathbb{E}[S]$ by -1.4% to $+0.1\%$, so it was removed.
- **More seeds at $N \leq 8$.** Eight more seeds per shape found exact circuits with fewer CZ for D_2^5 (9 against 10), D_3^6 (14 against 15) and D_3^7 (22 against 25), but none measured better on three calibrations: D_3^6 got worse (λ 0.8–1.0 against 0.25), and the other two tied within sampling noise while losing their robust angle picks. At small K a saved CZ is worth less than a lower λ; the teammates' 18-CZ D_3^7, seven CZ fewer, was the one that measured better. The same runs did shave bonus moments (D_2^5 15 $\to$ 13, D_3^7 23 $\to$ 21).
- **Shallower bonus synthesis.** Givens-only (excitation-preserving) brickwork, 2D patches instead of paths, and complex single-qubit layers with three times the parameters all needed as many layers as R_y+CZ on a path, or more. Divide-and-conquer beyond two arms needs $O(k)$ controlled rotations per thermometer bit, about as much as the sweep it replaces at $N \leq 15$.

References

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